



USB LED Light Cum Battery Charger

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Nowadays, many devices with rechargeable batteries use USB power to recharge them while they are connected. The USB interface has the ability to power devices, which can be used to charge a battery.

Here is an interesting yet simple circuit of a compact USB LED flashlight with built-in lithium-ion (Li-ion) battery and a dedicated battery-charger circuit. It charges the battery in a clever manner whenever the device is

docked to USB. The light source is a standard 1W white LED. Fig. 1 shows the author's prototype.

Circuit and working

Fig. 2 shows the circuit diagram of USB LED flashlight cum charger whose front end (charger) is wired around the advanced Li-ion linear charge-management chip MCP73831 (IC1). The 8-pin IC has features like high-accuracy current and voltage regulation, battery conditioning, charge termination and charge status indication. IC1 is followed by a commonly-available 3.7V/3.9Wh mobile phone Li-ion battery and the 1W white LED (LED3) with a simple MOSFET-based driver (T1).

The charging power supply is derived from the USB and the battery charging is fully controlled by MCP73831. Glowing of LED1 indicates that the battery is



Fig. 1: Author's prototype

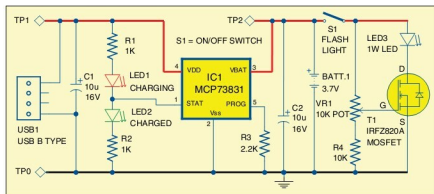


Fig. 2: Circuit diagram of USB LED flashlight cum battery charger

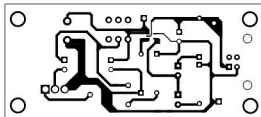


Fig. 3: PCB layout for the circuit

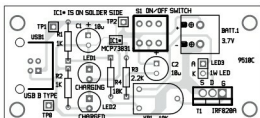


Fig. 4: Component layout for the PCB

PARTS LIST

Semiconductors:

- | | |
|---|--------------------------------|
| T1 | - MCF73831 Li-ion charger |
| T1 | - IRF2802A MOSFET |
| LED1, LED2 | - 5mm LED |
| LED3 | - 1W white LED |
| Resistors (all 1/4-watt, $\pm 5\%$ carbon): | |
| R1, R2 | - 1-kilo-ohm |
| R3 | - 2.2-kilo-ohm |
| R4 | - 10-kilo-ohm |
| VR1 | - 10-kilo-ohm linear potmeter |
| Capacitors: | |
| C1, C2 | - 10 μ F, 16V electrolytic |
| Miscellaneous: | |
| S1 | - On/off switch |
| BATT.1 | - 3.7V/3.9Wh Li-ion battery |
| USB1 | - USB B type connector |
| | - 2-pin terminal connector |

Test Points

Test point	Details
TP0	0V
TP1	5V
TP2	Battery voltage

charging and glowing of LED2 indicates that the battery is fully charged.

Switch S1 is used to turn flashlight LED3 on or off. Here, a potentiometer (VR1) is added deliberately to adjust the intensity of the light output from near 0% to 100% in a smooth manner. Note that, in this circuit configuration/application, a current-limiting resistor is not very crucial for the white LED3.

Construction and testing

An actual-size, single-side PCB for the circuit is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on a PCB, enclose it in a suitable plastic box. Use heat sink for LED3 or buy the IC with an embedded heat sink and connect it to the PCB using external wires. Before using the circuit, verify the test point voltages given in the table. ●

The author is a freelance technical writer and circuits designer